

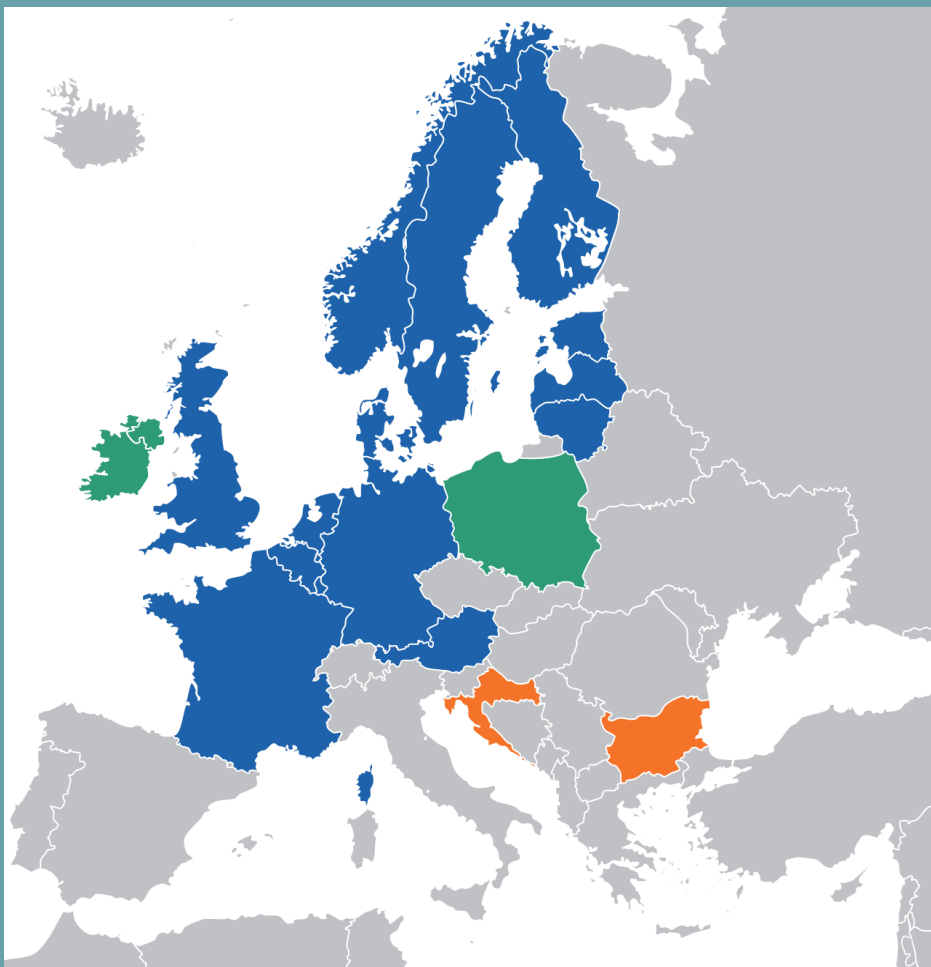
# FORECASTING FINNISH ELECTRICITY PRICES

SARIMAX-EGARCH & State Space Models

*by Martina Puglisi, Enio Hoxha and Franz Bittl*



THE TOPIC



# NORD POOL

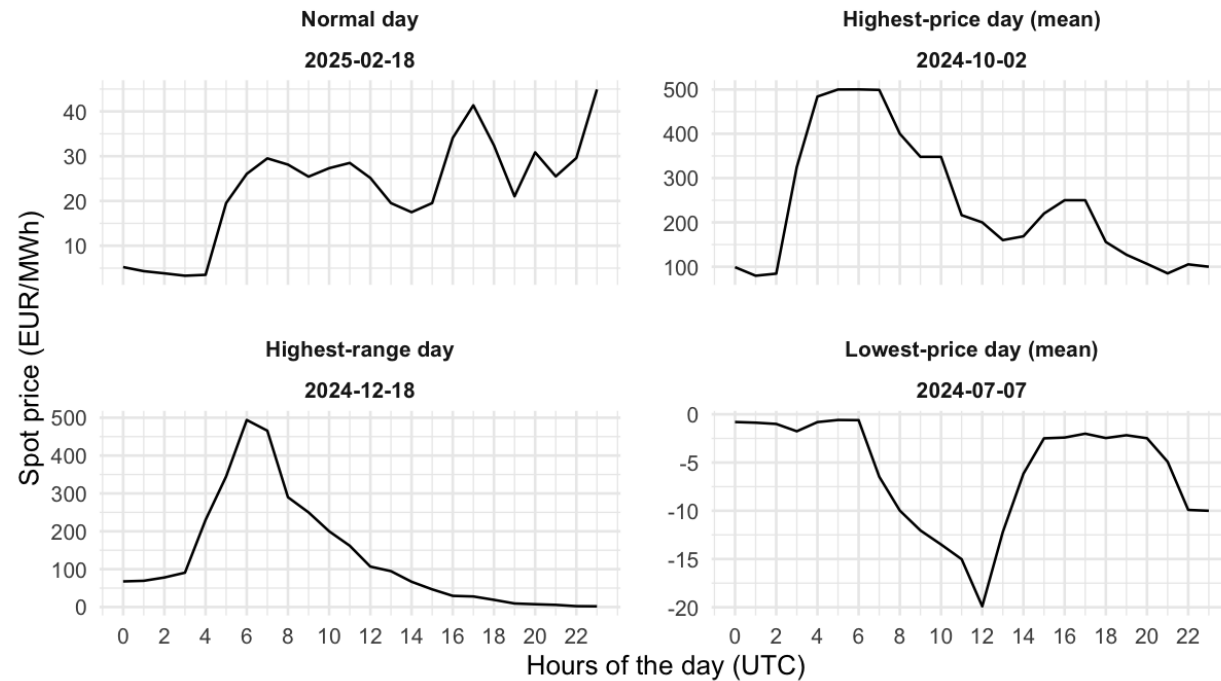
A EURONEXT COMPANY

# EXPLORING THE DATA

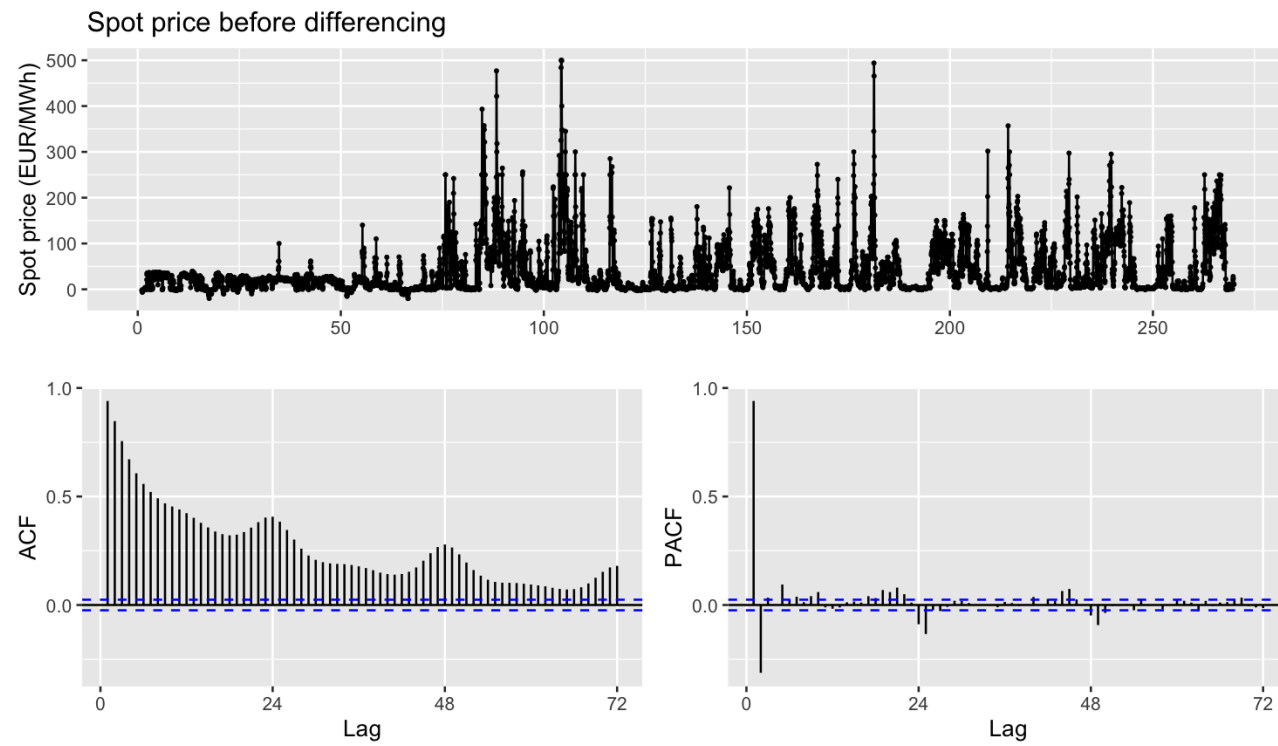


# Target variable

Hourly Spot Price Profiles on Selected Special Days

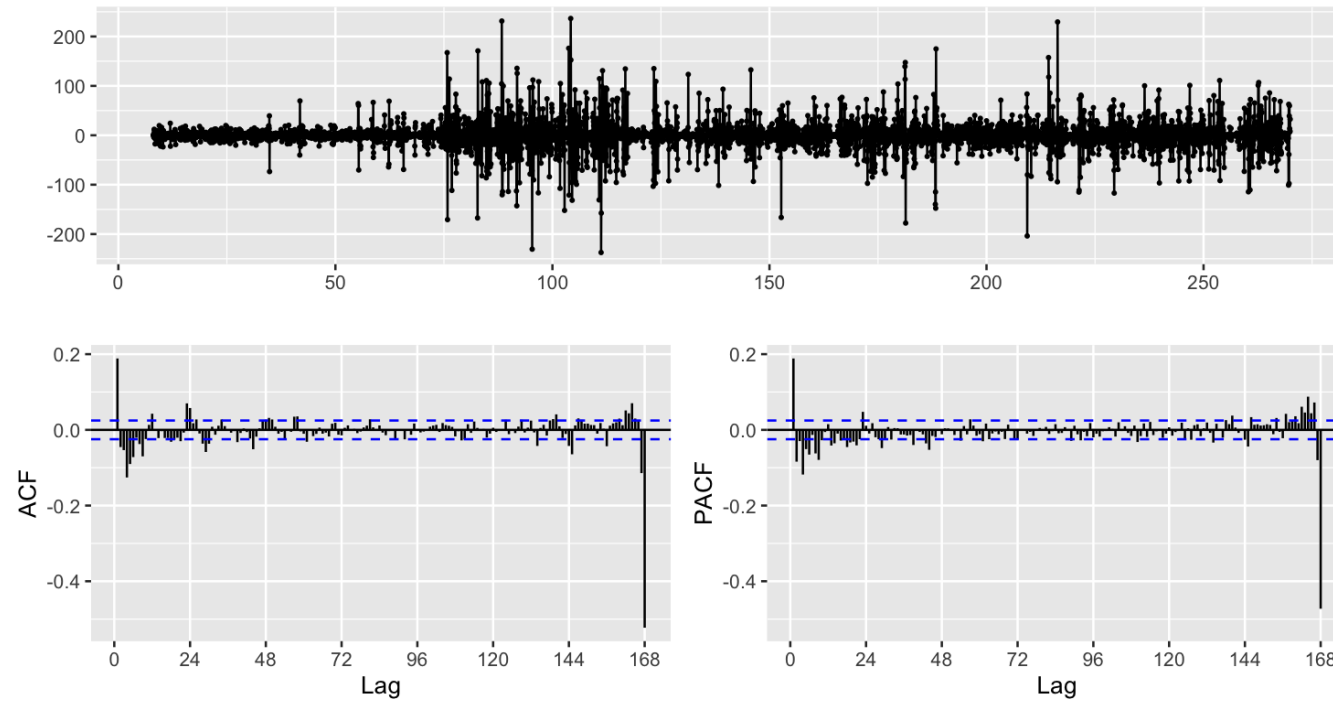


# Raw data



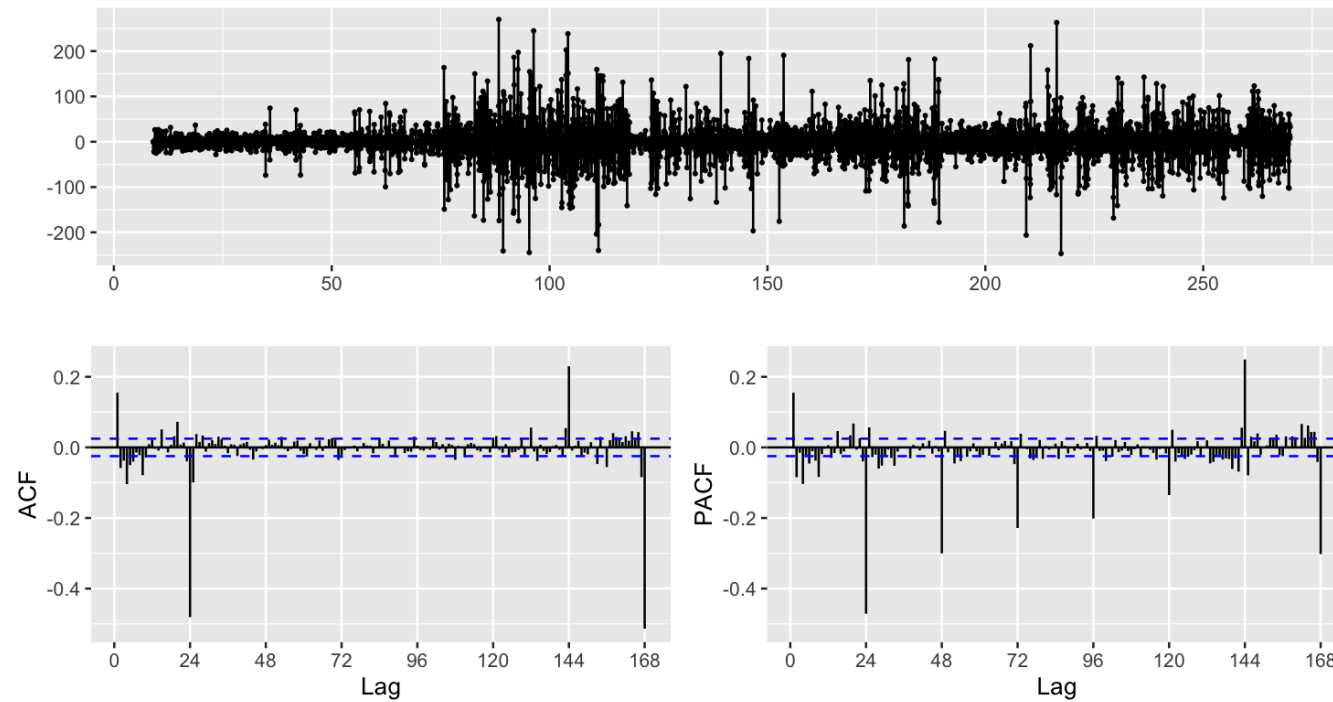
# Differenced data

ACF/PACF: First + Weekly (1, 168)



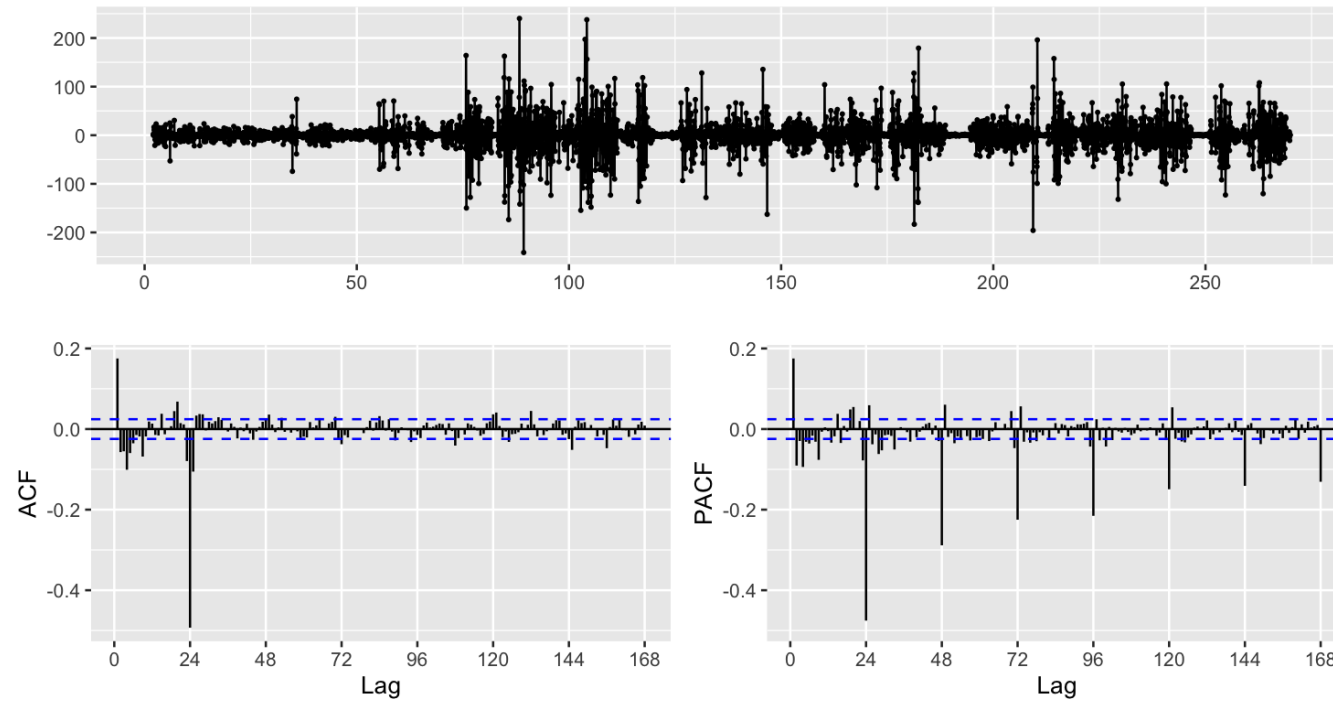
# Differenced data

ACF/PACF: First + Daily + Weekly (1, 24, 168)



# Differenced data

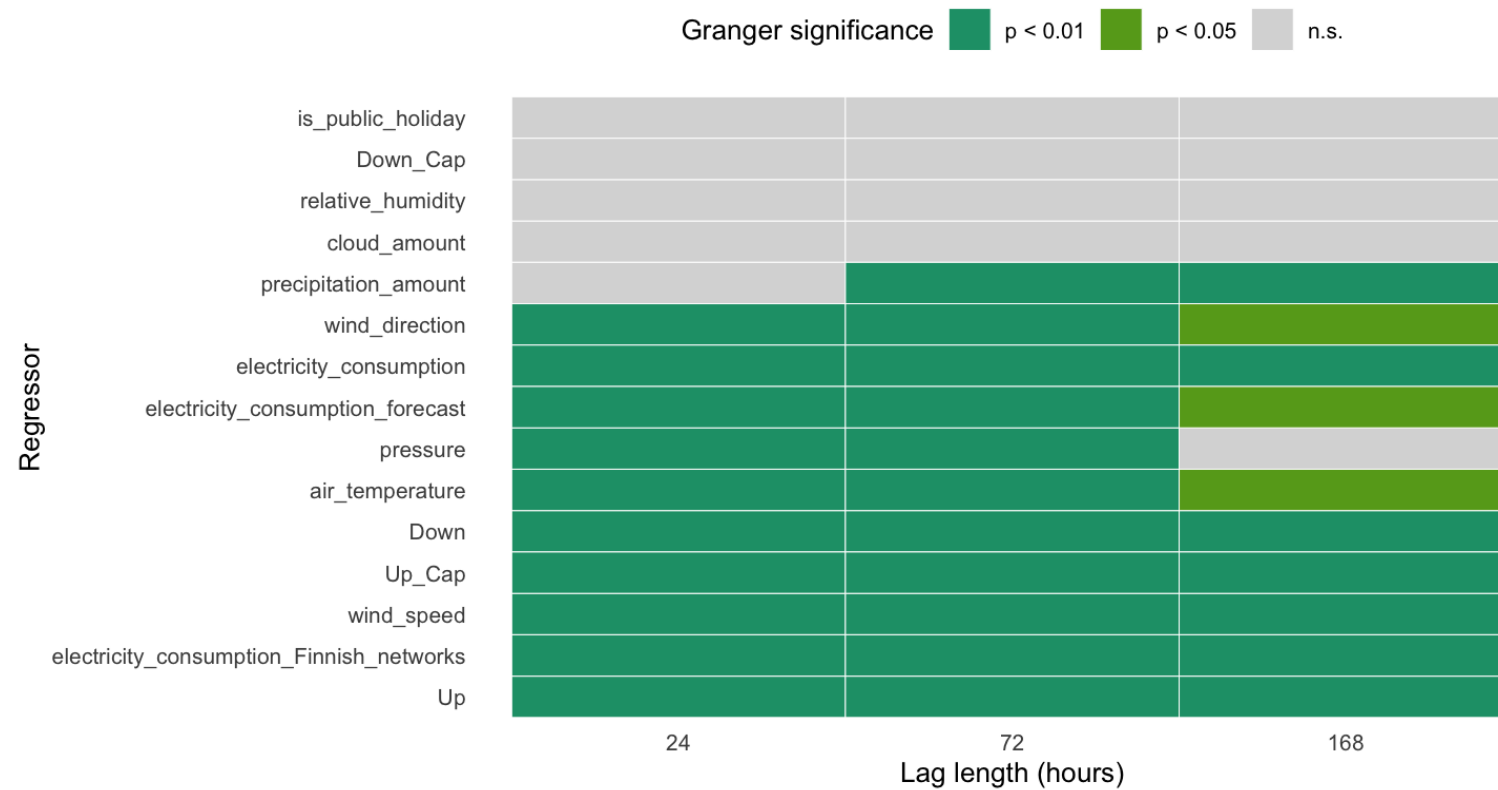
ACF/PACF: First + Daily (1, 24)



# Regressors

## Granger causality test across lag lengths (Normalized Variables)

Differenced series:  $\Delta(\text{normalized } X) \rightarrow \Delta(\text{normalized } sp)$



# SARIMAX-EGARCH



# Choosing the model: SARIMA

|    | Model_Name  | Specification  | AIC      | BIC      |
|----|-------------|----------------|----------|----------|
| M1 | (Benchmark) | (2,1,1)(1,0,0) | 53590.52 | 53637.74 |
| M2 | (Alt 1)     | (1,1,3)(0,1,1) | 52994.94 | 53048.87 |
| M3 | (Alt 2)     | (0,1,3)(0,1,1) | 53264.59 | 53311.78 |
| M4 | (Alt 3)     | (1,1,2)(0,1,1) | 52992.94 | 53040.13 |

Best Model:

SARIMA (1,1,2) (0,1,1) [24]

# SARIMAX: Test of coefficients

z test of coefficients:

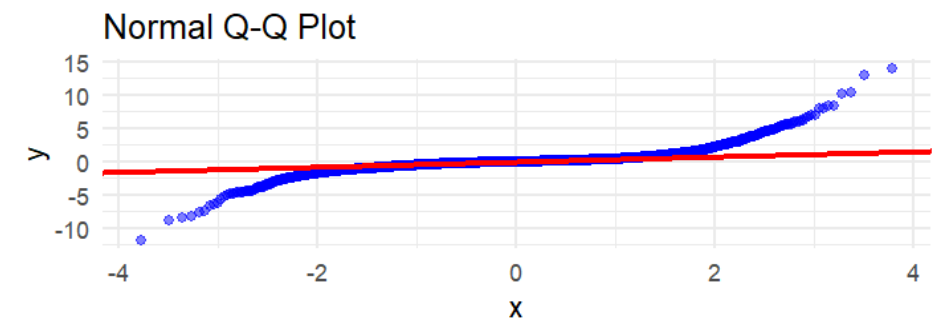
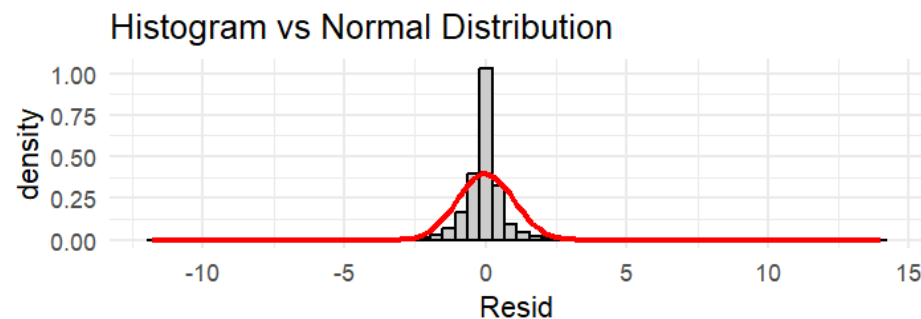
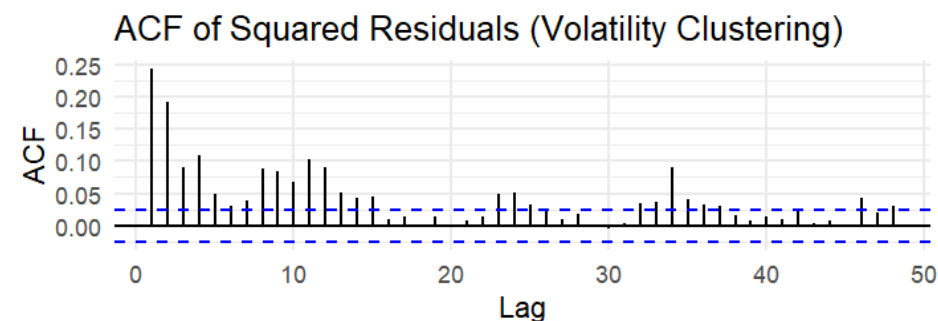
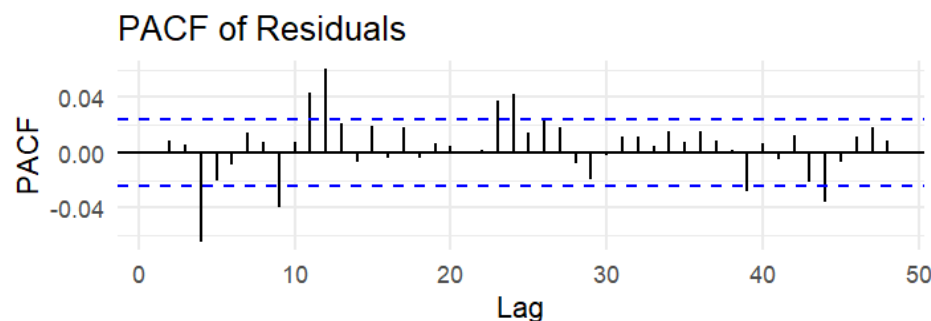
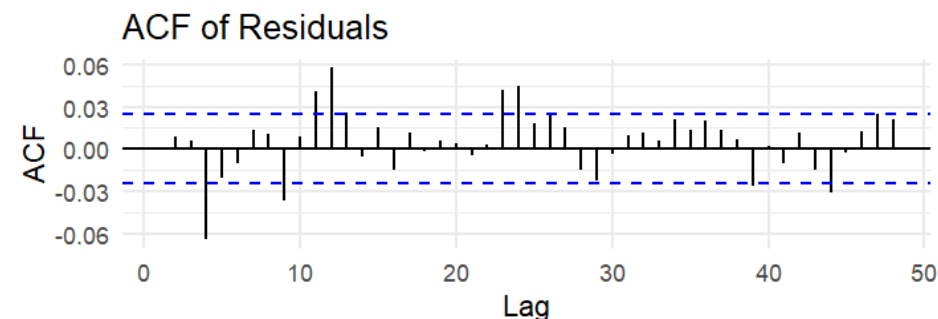
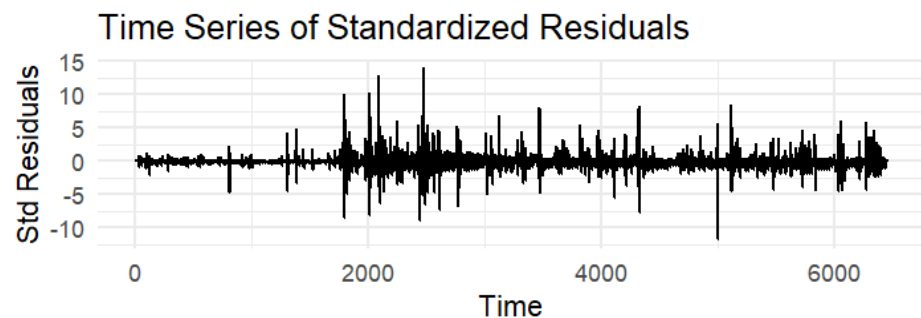
|          | Estimate   | Std. Error | z value   | Pr(> z )  |     |
|----------|------------|------------|-----------|-----------|-----|
| ar1      | 0.8953812  | 0.0104788  | 85.4468   | < 2.2e-16 | *** |
| ma1      | -0.7368844 | 0.0150958  | -48.8139  | < 2.2e-16 | *** |
| ma2      | -0.2571896 | 0.0130793  | -19.6638  | < 2.2e-16 | *** |
| sma1     | -0.9524950 | 0.0059025  | -161.3719 | < 2.2e-16 | *** |
| wind_std | -8.6985927 | 1.1912544  | -7.3020   | 2.834e-13 | *** |
| load_std | 0.4970838  | 0.8101848  | 0.6135    | 0.5395    |     |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

# SARIMA Diagnostic Analysis

Diagnostic Analysis: SARIMA(1,1,2)(0,1,1)[24]



# Choosing the model: GARCHX to eGARCH

|        | Estimate     | St.Error     | t value      | Pr(>  t )    |
|--------|--------------|--------------|--------------|--------------|
| omega  | 1.238742e+01 | 6.3786834330 | 1.942003e+00 | 5.213673e-02 |
| alpha1 | 1.000000e+00 | 0.2320468676 | 4.309474e+00 | 1.636432e-05 |
| beta1  | 3.799482e-01 | 0.0866115731 | 4.386806e+00 | 1.150272e-05 |
| vxreg1 | 9.940144e-09 | 5.4097040148 | 1.837465e-09 | 1.000000e+00 |
| vxreg2 | 9.195583e-09 | 1.0725545826 | 8.573533e-09 | 1.000000e+00 |
| vxreg3 | 1.004795e-08 | 4.3041896044 | 2.334458e-09 | 1.000000e+00 |

| Model                    | AIC    | BIC    | Persistence | Kurtosis |
|--------------------------|--------|--------|-------------|----------|
| -----                    | -----  | -----  | -----       | -----    |
| Model A (sGARCH-Norm)    | 7.7593 | 7.7624 | 0.9990      | 27.9800  |
| Model B (eGARCH-Norm)    | 7.6673 | 7.6715 | 0.9562      | 22.9781  |
| Model C (eGARCH-Student) | 7.2485 | 7.2537 | 0.8467      | 36.1313  |

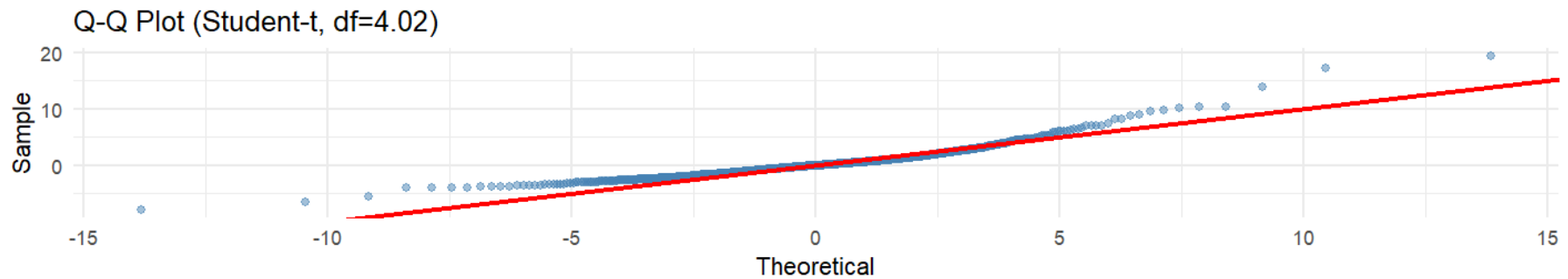
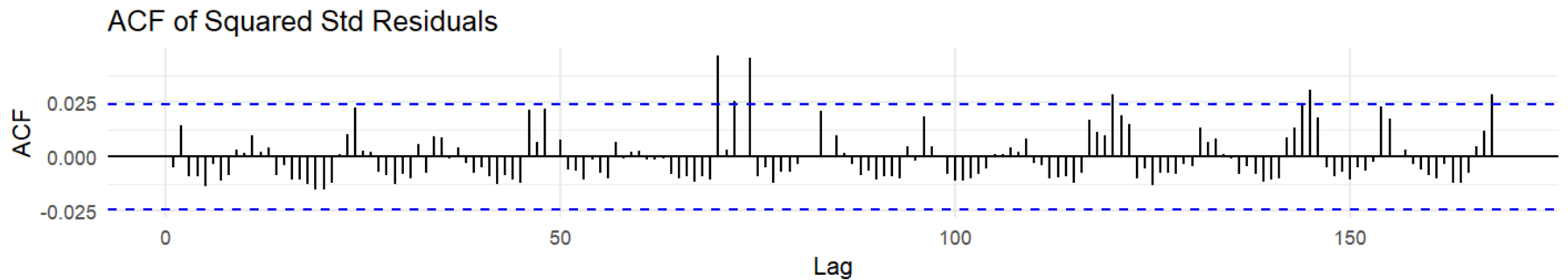
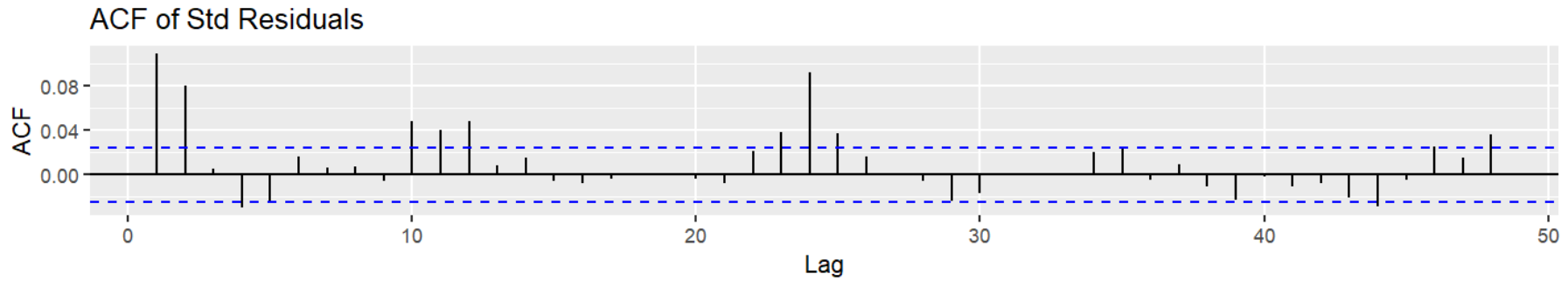
# Choosing the model: eGARCH

```
>>> DIAGNOSTIC eGARCH (Variance) <<<
      omega      alpha1      beta1      gamma1      shape
0.1038617 0.3797492 0.9924066 1.8441313 2.1000000
Ljung-Box (Residuals Std - Mean Dynamics): [1] 0.2190314
Ljung-Box (Residuals Std Quad - Vol Dynamics): [1] 0.9990991
```

ARCH-LM Test (Lag 24): p-value = 0.8783

The eGARCH model has successfully removed the volatility clustering.

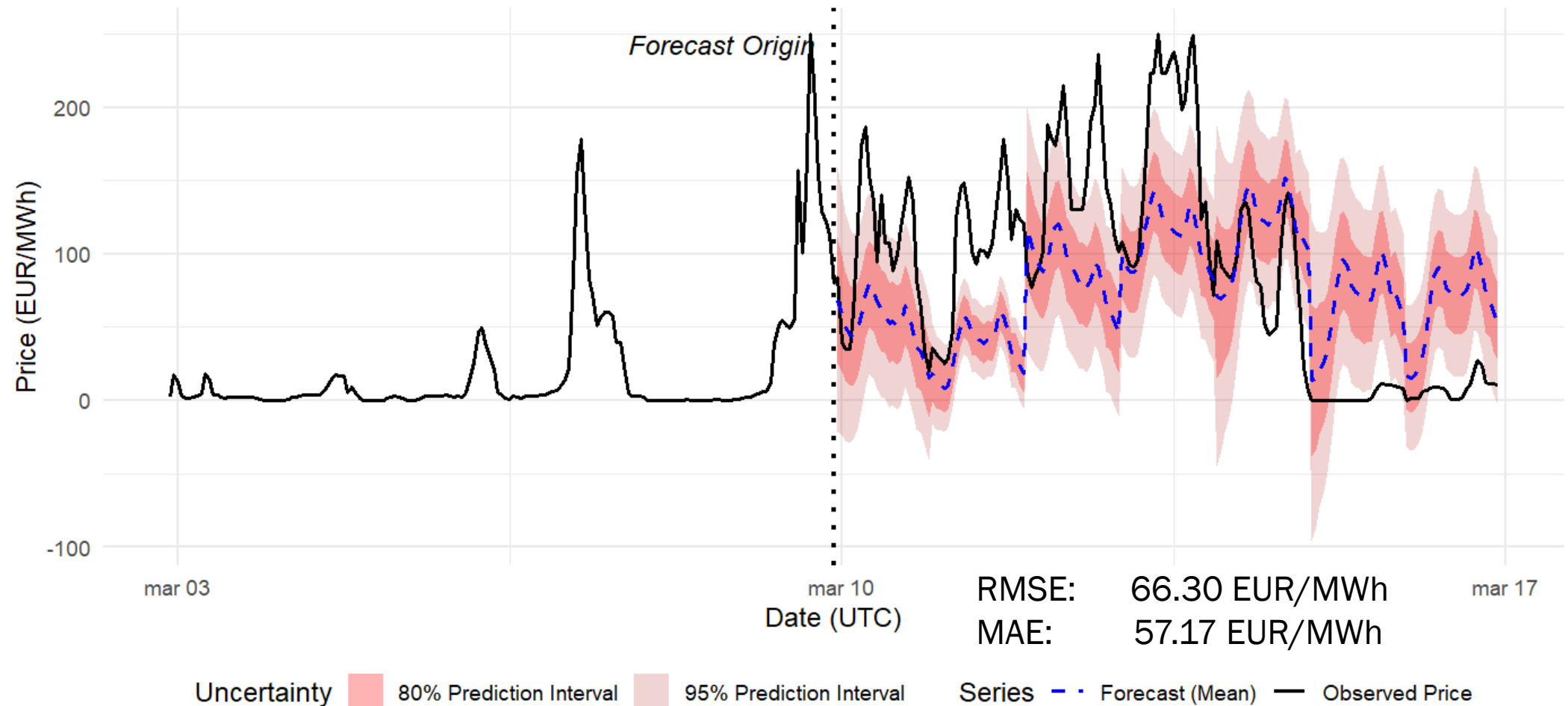
# eGARCH Diagnostic



# Pseudo Real-Time Forecasting

## Pseudo Real-Time Forecasting: SARIMAX + eGARCH-t

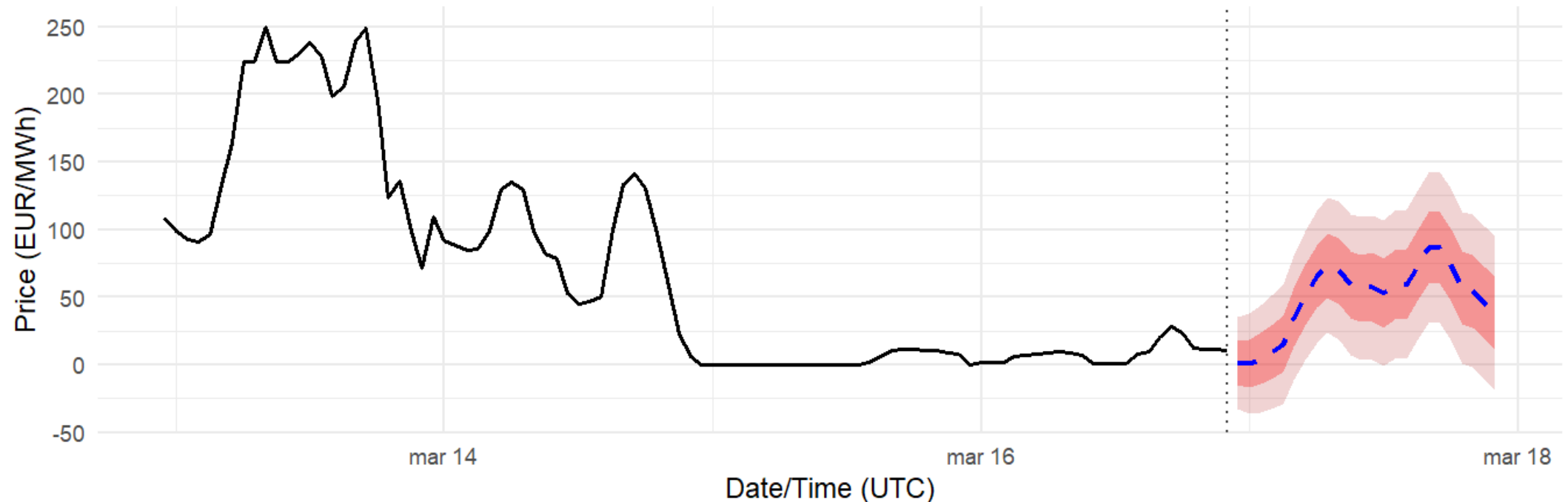
Evaluation of 1-Day Ahead Forecasts (Expanding Window Strategy)



# Forecasting out of sample 17 March 2025

## Out-of-Sample Forecast: March 17, 2025

Model: SARIMAX(1,1,2)(0,1,1) + eGARCH(1,1)-t (Wind Only)



Uncertainty ■ 80% Confidence ■ 95% Confidence Series - - Forecast (Mean) — Observed Price

Forecast conditional on persistent wind conditions from March 16.

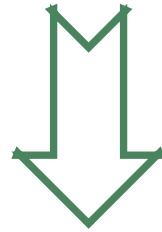
# STATE SPACE MODELS



# Why SSM for electricity prices

## OLS Models

Assuming the relationship between the electricity prices and the weather factors to **be constant** is simplistic.



## State Space Models

Allow trend, seasonality and regressors to be latent components that **evolve over time**.

# State Space Formulation

- Observation Equation – *electricity prices*

$$y_t = \mu_t + \gamma_t + \beta_{w,t} wind_t + \beta_{c,t} cons_t + \varepsilon_t \quad , \quad \varepsilon_t \sim N(\mathbf{0}, H)$$

- State Equation – *latent dynamics*

- Level – *trend*  $\mu_t = \mu_{t-1} + \eta_t \quad , \quad \eta_t \sim N(\mathbf{0}, Q_\mu)$

- 24h seasonality  $\gamma_t = \text{seasonal state evolution, variance } Q_\gamma$

- Random Walks – *time-varying betas*  $\beta_t = \beta_{t-1} + \xi_t \quad , \quad \xi_t \sim N(\mathbf{0}, Q_\beta)$

# Kalman Filter Forecasting Steps



1-day ahead latent states prediction

Using only information up  $t-1$ , the model predicts latent components at time  $t$ .



1-day ahead price prediction

Using 1-day ahead latent states prediction to forecast price.



OOS Rolling Week  
10th-16th March

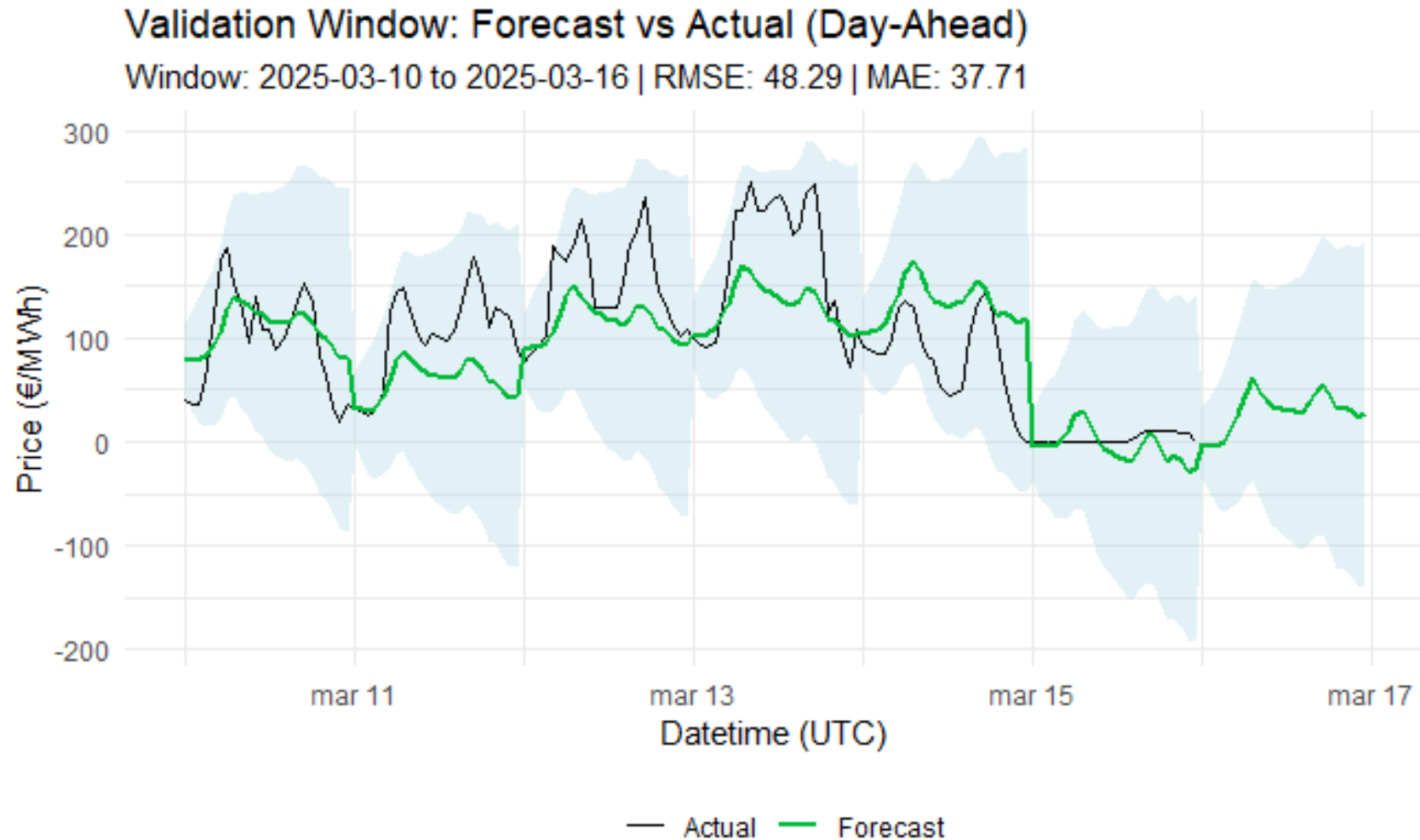
Forecast test is evaluated by RMSE and MAE.



17th March ahead  
Forecast

24-hour horizon spot price forecast.

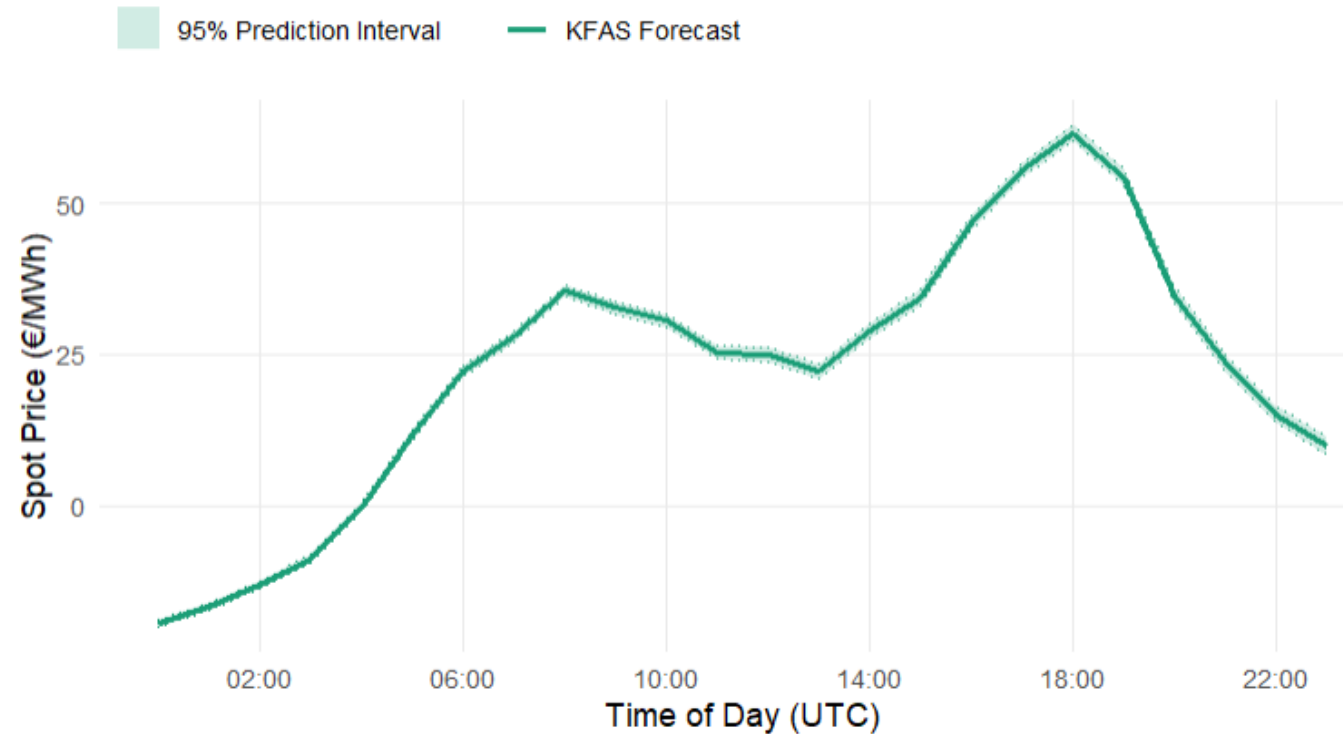
# Rolling out of sample evaluation



# 1-day ahead forecast for March 17th

## Day-Ahead Forecast: mar 17, 2025

Projected Spot Prices with 95% Confidence Bounds

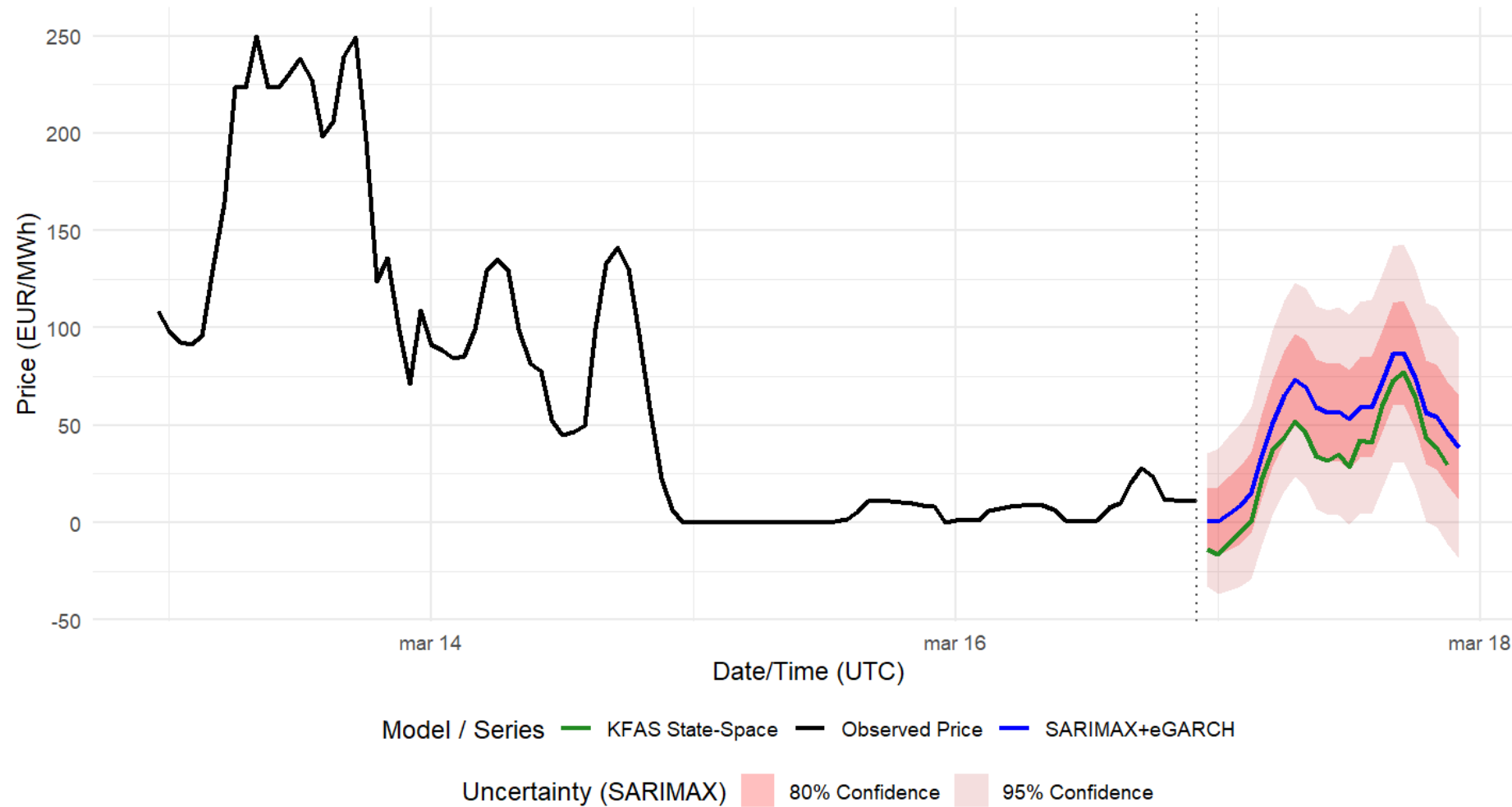


# 1-DAY AHEAD FORECASTING

SARIMAX-EGARCH vs. State Space Model

## Forecast Comparison: March 17, 2025

Econometric (SARIMAX) vs State-Space (KFAS) Approaches



A serene night scene of a snowy forest. The ground and trees are covered in a thick layer of white snow. The trees are dark evergreens, heavily laden with snow. In the background, the aurora borealis is visible as a vibrant green and blue light display against a dark, starry sky. The overall atmosphere is peaceful and cold.

THANK YOU FOR YOUR ATTENTION

MEF LAB